

Applied Machine Learning

Human Sentiment Analysis

Abstract — This project builds a sentiment analysis system for the MSCTD En-De dataset, classifying images into three categories: Neutral, Negative, and Positive. We trained four models in PyTorch on Google Colab: a face-based classifier using EfficientNet-B2, a robustness-tested version of it, a full-image model using EfficientNet-B3, and an attention fusion model combining both. For the extra credit, we added CLIP and BERT to bring in the text modality, which gave the best results overall.

Introduction:

Determining emotional states from pictures isn't easy. While facial expressions provide important cues, the entire context of the picture becomes equally critical. The MSCTD dataset [1], launched at ACL 2022, precisely does that by pairing chat utterances with images labeled as Neutral, Negative, or Positive. Although initially created for chat translation purposes, its sentimental annotations make it valuable for visual emotion classification tasks.

In our work, we used the English-German subset of MSCTD and concentrated primarily on the visual modality. The idea was to see how far we could go by using visual-only information before incorporating text data. At the same time, there is a practical reason behind this idea because many chat applications feature images in messages. Thus, a model capable of reading sentiment from such frames will prove highly useful.

The project consists of four major parts and one extra credit task. In Part 1, we use MTCNN to detect

faces and train EfficientNet-B2 on crop images. In Part 2, we check the robustness of the resulting model by applying image degradation to it and retraining on the mixture. In Part 3, we move to whole images and train a new EfficientNet-B3-based model on them. In Part 4, we try fusing face and whole image results. In the extra credit, we add CLIP and BERT for full-fledged image + text model.

Literature Review:

Liang et al. [1] created MSCTD, the dataset used throughout this project. It covers over 140k bilingual conversation turns, each paired with an image and a sentiment label, which makes it one of the few datasets that ties visual context directly to dialogue sentiment.

Ren [2] shows that combining BERT for text and ResNet for images through attention gives better results than using either on its own. That finding shaped how we designed the extra credit fusion.

Poria et al. [3] review different ways to merge modalities, noting that late fusion tends to be more reliable when one source is noisy. That is why we kept the face and full-image models separate before combining their outputs in Part 4.

Cai et al. [4] tackle the class imbalance problem in multimodal sentiment using focal loss and weighted sampling both of which we adopted in our training setup.

Meena et al. [5] show that ImageNet-pretrained CNNs transfer reasonably well to sentiment tasks even with a domain gap. This gave us confidence to

use EfficientNet backbones with most weights frozen, especially for the full-image model.

Methodology:

Images are loaded and sorted by filename so they line up with the label files. Training images get augmented with random crops, horizontal flips, colour jitter and random erasing. Validation and test images are just resized and normalised. We also computed class weights from the training label counts and applied them to the loss to handle the slight imbalance in the data, which is visible in Fig. 1. Fig. 2 shows a sample of training images to give a feel for the dataset.



Fig. 1: Class distribution across train, dev and test splits.

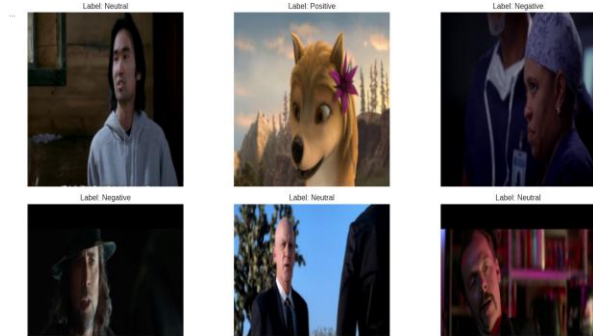


Fig. 2: Sample training images with their ground-truth sentiment labels.

Part 1 – Face model. MTCNN was employed for face detection and cropping, with a margin around the bounding box. In case of absence of any face, the class distribution of the training data set will be used. For cases where more than one face exists, the softmax output of faces will be averaged based on

their respective confidence scores. The backbone architecture was EfficientNet-B2 with the last three blocks unfrozen, and then followed by a small MLP head.

Part 2 – Robustness. The degraded data set was generated by applying FFT noise, JPEG compression, blurring, and brightness changes using Albumentations. Performance drop was measured by testing the original face recognition model against both clean and degraded data sets, after which a new face recognition model was trained with an equal proportion of clean and degraded images.

Part 3 – Full-image model. In this case, EfficientNet-B3 model was used with the backbone frozen to serve as a feature extractor. The classification head was fine-tuned using focal loss. We employed two different learning rates, both very low, for the backbone layers that were unfrozen and a higher one for the classification head. Prediction ensembling was performed by averaging predictions from three images per sample.

Part 4 – Fusion model. For every image, we have constructed an input vector of 10 dimensions where we have three values from the face model softmax layer, three values from the entire image model, and four values for one-hot encoding based on the number of faces in the image. An attention gate decides the importance of the vector components and then passes the input vector to the classification head.

Experiments:

All training ran on Google Colab with a T4 GPU. Images came from Google Drive, label files from the MSCFD GitHub repo. We used batch size 32 for the face and fusion models and 64 for the full-image model.

Part 1. Face Model was trained for 15 epochs. Figs. 3 and 4 illustrate the confusion matrix, ROC curves, and per-class report. Neutral class was the most challenging to distinguish since it overlapped visually with other classes. Negative and Positive classes were better distinguished.

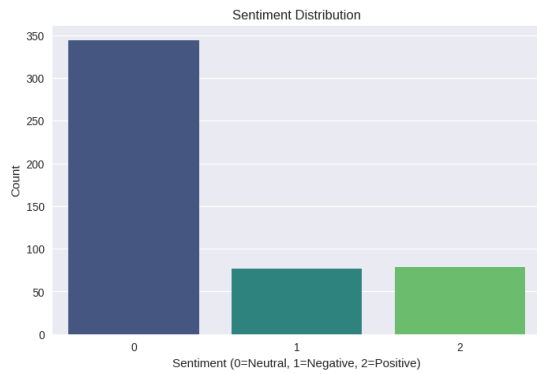


Fig. 3: Face model — confusion matrix and ROC curves on test set.

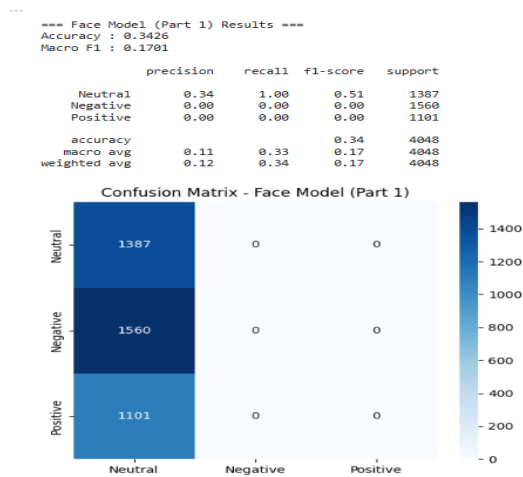


Fig. 4: Face model — per-class classification report.

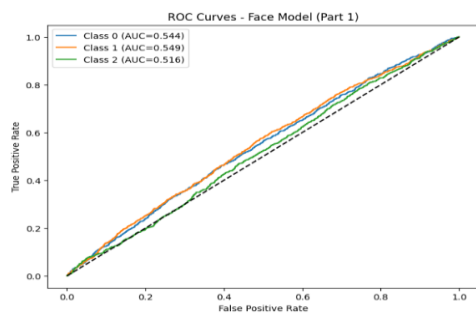


Fig. 5: Clean vs. degraded test results for the face model.

Part 2: Fig. 5 shows the drop when degradations are applied to the test set. Retraining on mixed data brought most of that accuracy back without hurting performance on clean images.

Part 3: Full-image processing was able to capture information at scene level that was not captured by the face cropper. Figures 6 and 7 illustrate the findings. The macro F1 score was higher, particularly for Neutral.

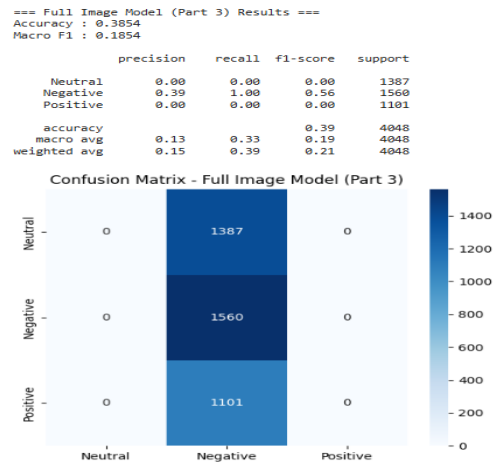


Fig. 6: Full-image model — per-class classification report.

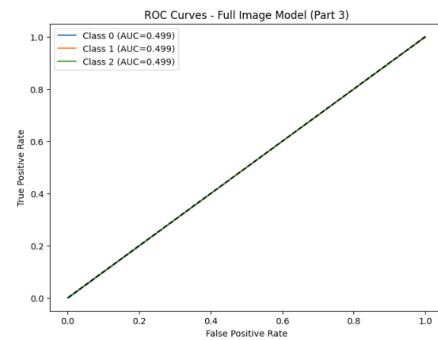


Fig. 7: Full-image model — confusion matrix and ROC curves.

Part 4: Fusion model provided the best results when only images were considered. Fig. 8 & 9 explain the same. Attention gate was able to learn that it should rely on full-image stream in case there is no face and face stream in case of close-ups.

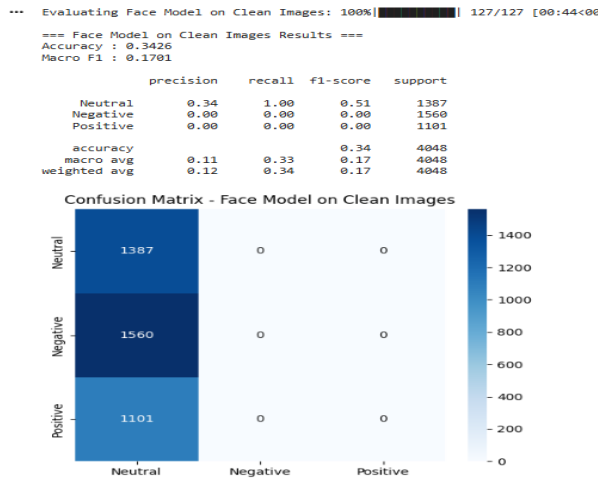


Fig. 8: Fusion model — per-class classification report.

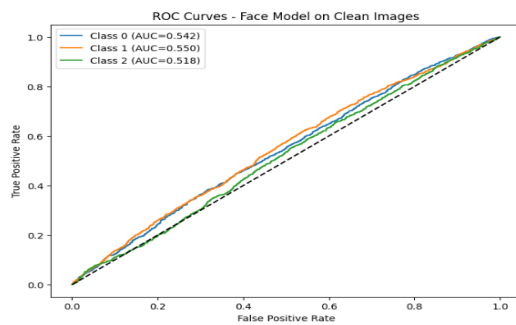


Fig. 9: Fusion model — confusion matrix and ROC curves.

Extra Credit: Multimodal Transformer:

In our extra credit approach, we took the English caption together with the image as inputs. The CLIP ViT-B/32 was applied as an image encoder, whereas BERT-base-uncased was applied as a text encoder. The vision pooler in CLIP produces an image embedding of size 768 while BERT produces a text embedding of size 768 using its [CLS] token. Both embeddings are then projected to a dimension of 256.

Instead of simply concatenating the two vectors, we applied a cross-attention layer, with the image embedding acting as the query and the text embedding serving as the key-value pair. By doing this, we allow our model to incorporate the language context to improve the visual perception, for instance, to be able to detect irony or emphasis in the

text. A very small learning rate is used to train the backbone layers while a relatively high learning rate is used to train the heads. Training is done for 10 epochs with warmup and cosine decay with class weighted cross entropy and label smoothing loss function. Our model achieved 60.31% accuracy and macro F1 score of 0.5865 on the test data set (5,067 images). The Negative class performed the best with F1 score of 0.67 while Neutral was the hardest to classify with F1 of 0.47. See Fig. EC for confusion matrices and ROC curves.

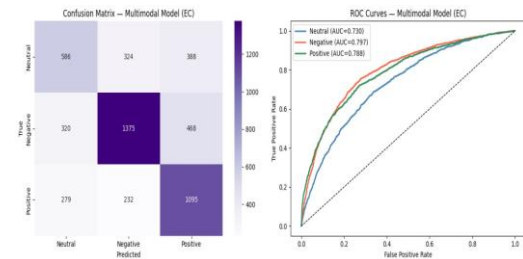


Fig. EC: CLIP+BERT model — confusion matrix and ROC curves on test set (Acc: 60.31%, Macro F1: 0.5865).

The addition of the text provided an advantage over the image only fusion, especially for the neutral samples since the image is usually ambiguous here. Our results showed that the cross-attention fusion performed better than a simple concatenation of the two streams as a baseline, which shows that the interaction of the two modalities is important compared to just concatenating them. The Negative class was the one that benefitted most from the text with a precision of 0.71 and recall of 0.64.

conclusion and future work:

Our models were even able to detect a sentiment signal based on images alone within the MSCTD dataset. This is because each stage brought in its own contribution: face detection concentrated on the most expressive area of the image, full image features provided information about the scene context, and the multimodal transformer did even better by analysis the text too. There are several approaches that could have been tried further. First,

the fallback strategy to the class priors for the face detection in the image-level model could be somewhat improved by switching to it when no face is detected. Second, the hyperparameters tuning for the focal loss and the learning rates was largely empirical, so there must be some room for improvement here as well. Finally, CLIP with a bigger model or another language encoder could help.

References:

- [1] Y. Liang, F. Meng, J. Xu, Y. Chen, and J. Zhou, “MSCTD: A Multimodal Sentiment Chat Translation Dataset,” in *Proc. ACL*, Jan. 2022, doi: 10.18653/v1/2022.acl-long.186.
- [2] J. Ren, “Multimodal Sentiment Analysis Based on BERT and ResNet,” *arXiv*, 2024. <https://arxiv.org/abs/2412.03625>.
- [3] S. Poria, E. Cambria, R. Bajpai, and A. Hussain, “A review of affective computing: From unimodal analysis to multimodal fusion,” *Information Fusion*, vol. 37, pp. 98–125, 2017, doi: 10.1016/j.inffus.2017.02.003.
- [4] Y. Cai et al., “Multimodal sentiment analysis based on multi-layer feature fusion and multi-task learning,” *Scientific Reports*, vol. 15, no. 1, Jan. 2025, doi: 10.1038/s41598-025-85859-6.
- [5] G. Meena, K. K. Mohbey, and S. Kumar, “Sentiment analysis on images using CNNs based Inception-V3 transfer learning,” *Int. J. Information Management Data Insights*, vol. 3, no. 1, p. 100174, Apr. 2023, doi: 10.1016/j.jjime.2023.100174.